

Pack a Lunchbox SOP (3x)

Pack three lunchboxes in one session using a tabletop robot with two arms. Record each lunchbox as its own episode, for a total of three episodes per session. In each episode, one set of items runs from the left staging plate into the open lunchbox base in the center, the base is turned a quarter turn, its detachable lid is fitted and latched, and the closed box is placed into one lunch bag in the right output zone.

Each episode packs one three-compartment rigid lunchbox with a **detachable four-latch lid** — one latch centered on each of the four edges. The sandwich arrives **already wrapped**, so there is no wrapping in this task. The item set per box is: one **wrapped sandwich** (a half sandwich, wrapped before the episode), one whole fruit, one packaged snack, and one folded paper napkin — all staged on the left plate. The lid carries its own utensil in a molded channel on its top, so the utensil is not a loose item and is not packed. Compartment assignments are fixed: the wrapped sandwich goes in the large compartment **C1**, the fruit in the upper-right compartment **C2**, the snack in the lower-right compartment **C3**, and the napkin flat over C1. Handle food gently: never crush the sandwich, bruise the fruit, or dent the snack packaging.

Setup

Complete both checklists before the first episode of the session. Before episodes 2–3, repeat the Hardware checklist and confirm the remaining item sets, lunchbox bases, lids, and bags are in their expected states.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. Env camera frame includes the front and back table edges, centered on the table midpoint, with the left staging plate, the center lunchbox base, the lid, and the right output zone (lunch bag) all in frame.
3. Both arms are at the home position with grippers open.
4. Table is clear of everything except the current item set on its plate, the open lunchbox base, the detachable lid, and the lunch bag.

Materials checklist

1. One **item set** on the **left** staging plate, items separated and not overlapping: one **wrapped sandwich** (about 12 × 6 × 6 cm, a pre-wrapped half sandwich), one whole fruit, one

packaged snack, and one folded napkin. Confirm the sandwich is fully wrapped with no bread or filling exposed; if it is not, treat it as defective material and replace it.

2. One **open lunchbox base** in the center working area, base flat on the table, compartments empty. The base has no lid attached — the lid is staged separately.
3. One **detachable lid** staged flat on the **right**, just right of the base, with its **utensil** seated in the molded **utensil channel** on its top and both faces clear.
4. One **open lunch bag** upright in the right output zone, beyond the lid, base down, mouth held open, tall enough that the closed lunchbox sits with its top edge at least 2 cm below the bag opening.
5. Before the session, confirm the wrapped sandwich fits fully inside C1 below the rim, the fruit fits in C2 below the rim, the snack fits in C3 below the rim, the lid seats flat on the filled base, and the closed, latched lunchbox drops flat to the bottom of the bag.

Workspace layout

- **Left:** staging plate holding the four items for the current episode (wrapped sandwich, fruit, snack, napkin)
- **Center:** open lunchbox base
- **Right:** the detachable lid staged just right of the base, and the open lunch bag beyond it in the output zone

Vocabulary

- **Item set:** the four items packed per episode — wrapped sandwich, fruit, snack, and napkin. One set per box, staged on the left plate. The utensil is not part of the set; it rides in the lid's top channel.
- **Wrapped sandwich:** a half sandwich wrapped before the episode. It is packed as-is; there is no wrapping step. Called "the sandwich" in the steps.
- **Detachable lid:** the separate lid that lifts fully off the base. It carries the utensil in a molded channel on its top and fastens with four latches.
- **Compartments:** **C1** large (left two-thirds), **C2** upper-right, **C3** lower-right.
- **Utensil channel:** the molded channel on the top of the lid that holds the utensil. It comes pre-loaded and is not packed during the episode.
- **Filled base:** the lunchbox base after the sandwich, fruit, snack, and napkin are seated, before the lid is fitted.
- **Seated below the rim:** an item's top surface is at or below the compartment rim (proud by no more than 5 mm), so the lid seats flat.
- **Proud of the rim:** an item's top rides above the rim; the lid cannot seat flat.
- **Latched:** all four lid latches engaged with a lid-to-base gap under 5 mm all around.

- **Seated in the bag:** the closed, latched lunchbox rests flat on the bag bottom, base down and upright, with its top edge at least 2 cm below the bag opening.

Steps

Step 1: Place the wrapped sandwich in C1

Goal: the wrapped sandwich rests flat in C1, seated below the rim.

- With the **left gripper**, lift the wrapped sandwich from the left plate.
- Lower it flat into compartment **C1**, any loose or uneven folded edges underneath.
- Release when it sits flat and does not spring up.

Check: the wrapped sandwich is fully inside C1 with its top surface at or below the rim (proud by no more than 5 mm). If it sits proud of the rim, lift it, press it flatter, and reseal.

Step 2: Add the fruit to C2

Goal: the fruit rests in C2, seated below the rim and not rolling.

- With the **left gripper**, lift the fruit from the left plate.
- Lower it into compartment **C2**.
- Release only when the fruit is stable and not rolling.

Check: the fruit is fully inside C2, below the rim, and not touching the sandwich in C1. If it rides above the rim or crosses into C1, reseal it lower in C2.

Step 3: Add the snack to C3

Goal: the snack rests flat in C3, seated below the rim.

- With the **left gripper**, lift the packaged snack from the left plate by its packaging edge without denting it.
- Lay it flat into compartment **C3**.
- Release when it lies flat below the rim.

Check: the snack is fully inside C3 with its top at or below the rim. If it stands on edge or rides proud of the rim, reseal it flat.

Step 4: Lay the napkin over C1

Goal: the folded napkin lies flat over the wrapped sandwich, inside the rim of C1.

- With the **left gripper**, lift the folded napkin from the left plate.
- Lay it flat and centered over compartment **C1**, on top of the wrapped sandwich, edges inside the rim.
- Press it flat with the **right gripper** so no corner rides above the rim.

Check: the napkin covers C1 with all edges inside the rim and no corner proud of the rim. If a corner rides up, refold or reseal that corner inside the rim.

Step 5: Rotate the filled base 90° clockwise

Goal: the filled base is turned a quarter turn clockwise and set back flat, with every item still seated.

- With **both grippers**, grasp the filled base on its left and right sides and lift it level, clear of the table, keeping it flat so nothing shifts.
- Rotate it **90° clockwise** (viewed from above).
- Lower it flat to the table in the center working area and release when it sits flat and square.

Check: the base sits flat and square, all items are still seated below their rims, and the napkin is still flat over C1. If the fruit rolled, the snack shifted, or the napkin slipped, reseal it (Steps 2–4) before fitting the lid.

Step 6: Fit and latch the lid

Goal: the lid is seated flat on the base and all four latches are engaged.

6.1 Fit the lid onto the base

- With the **right gripper**, lift the detachable lid (utensil pre-loaded in its top channel) from the right.
- Lower it squarely onto the base, utensil channel facing up, edges aligned with the base all around.
- Press it down flat with the **right gripper** until it meets the base with a gap under 5 mm all around.

6.2 Engage all four latches

- With the **left gripper**, hold the lid down; with the **right gripper**, close the **right** latch, then the **front** latch.
- With the **right gripper**, hold the lid down; with the **left gripper**, close the **left** latch, then the **back** latch.
- Press each latch until it seats.

Check: all four latches are engaged and the lid-to-base gap is under 5 mm all around. If the lid rides up, a latch pops, or the gap exceeds 5 mm, an item is proud of a compartment — unlatch, lift the lid, reseal the highest item below its rim (Steps 1–4), then refit and relatch.

Step 7: Bag the closed lunchbox

Goal: the closed, latched lunchbox sits flat on the bottom of the lunch bag, base down.

- With the **left gripper** and **right gripper**, grasp the closed lunchbox on its left and right sides and lift it level, base down, clear of the table.
- Carry it over the open lunch bag and lower it straight down, base first, to the bottom of the bag.
- Keep the box upright and square to the bag as it descends.
- Release when the box rests flat on the bag bottom.

Check: the lunchbox sits flat on the bag bottom, base down and upright, with its top edge at least 2 cm below the bag opening. If it lands tilted or wedged on the bag wall, lift it and lower it straight down again.

Step 8: End the episode

Goal: end the current episode after the box is packed, latched, and seated in the bag.

- Confirm the closed, latched lunchbox rests flat in the lunch bag.
- Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

These actions happen while data collection is stopped.

- **After episodes 1–2:** leave the packed lunchbox in its bag and move it aside. Confirm the center working area is clear, then stage a fresh item set on the left plate, a fresh open lunchbox base in the center, a fresh lid on the right, and a fresh open lunch bag in the output zone. Confirm both arms are home with grippers open. Start a new recording, then pack one lunchbox.
- **After episode 3:** confirm three packed, bagged lunchboxes are set aside. Reset the full session with recording off: clear the table, stage three fresh item sets on plates, three open lunchbox bases, three lids, and three open lunch bags for the next session, one set per episode. Confirm the table holds only the current episode's materials, then run both Setup checklists again.

SOP violations

Things that break the SOP. This list feeds the violation set and is what reviewers look for with the side by side review tool.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is what the reviewer sees in the video. The coaching note is for retraining and is not an annotation label.

Episode handling

Assign the same session ID to all three episodes and label each recording with its episode number (1, 2, or 3). Each episode is fully independent: one item set, one lunchbox base, one lid, one bag.

Tag every violation with its timestamp and name. Retain the episode in training data with the violation tag. Do not delete it solely because of a violation.

Violations

Violation: Wrong compartment or placement order

- **Visible cue:** an item goes in the wrong compartment (sandwich outside C1, fruit outside C2, snack outside C3, napkin not over C1), items overlap compartments, or the fixed order is not followed.
- **SOP rule broken:** Steps 1–4 (pack sandwich → C1, fruit → C2, snack → C3, napkin over C1, in that order).
- **Coaching note:** follow the fixed compartment map and packing order every time.

Violation: Item left proud of the rim

- **Visible cue:** an item's top rides above the lunchbox rim (more than 5 mm proud) and is covered by the lid anyway.
- **SOP rule broken:** Steps 1–4 checks (each item seated at or below its compartment rim before moving on).
- **Coaching note:** seat each item below the rim before adding the next.

Violation: Food handled roughly

- **Visible cue:** a gripper crushes the sandwich, bruises the fruit, dents the snack packaging, or drops an item into a compartment.
- **SOP rule broken:** Steps 1–4 (lift and lower each item gently; grasp packaging edges without denting).
- **Coaching note:** grasp gently, lower items in, do not drop or crush.

Violation: Base not rotated or items dislodged in the turn

- **Visible cue:** the filled base is lidded without the 90° clockwise turn, is turned the wrong way or a different amount, or an item or the napkin shifts, rolls, or falls during the turn and is not reseated before the lid goes on.
- **SOP rule broken:** Step 5 (lift the filled base level, turn it 90° clockwise, set it back flat, and reseal any item that shifts before fitting the lid).

- **Coaching note:** keep the base level through the turn and reseal anything that moves before lidding.

Violation: Lid not seated flat or latches not engaged

- **Visible cue:** the lid is set on crooked or proud, left off, put on with the utensil channel down, or one or more of the four latches is left open, or the lid-to-base gap exceeds 5 mm and the box is bagged anyway.
- **SOP rule broken:** Step 6 (fit the lid square, utensil channel up, and engage all four latches with a lid-to-base gap under 5 mm).
- **Coaching note:** seat the lid square with the utensil channel up, engage all four latches, and if a latch pops, reseal the tallest item before relatching.

Violation: Lunchbox not seated flat in the bag

- **Visible cue:** the box is left on the table, placed beside the bag, dropped in on its side or tilted, wedged on the bag wall, or left with its top above the bag opening.
- **SOP rule broken:** Step 7 (lower the closed box base-first, flat on the bag bottom, upright, top at least 2 cm below the bag opening).
- **Coaching note:** lower base-first and straight down until the box rests flat on the bag bottom.

Violation: Wrong arm used

- **Visible cue:** a step that names a gripper is done with the other one.
- **SOP rule broken:** Steps 1 to 7, each action uses the gripper its step names.
- **Coaching note:** walk the left/right roles through step by step.

Violation: Wrong episode ending

- **Visible cue:** data collection ends before the box is latched and bagged, or the arms fail to return home with grippers open after the box is seated in the bag.
- **SOP rule broken:** Step 8 (confirm the closed, latched box rests flat in the bag, then return both arms home with grippers open before ending data collection).
- **Coaching note:** bag the latched box, confirm it is seated, and home both arms before ending.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, or motor error.

- **Defective materials:** a sandwich that arrives unwrapped or with filling exposed, a spoiled or crushed food item, a broken latch or lid, or a bag that will not stay open. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Place the wrapped sandwich in compartment C1
2. Add the fruit to compartment C2
3. Add the snack to compartment C3
4. Lay the napkin over C1
5. Rotate the filled base 90° clockwise with both arms
6. Fit the lid and engage all four latches
7. Place the closed lunchbox flat into the lunch bag
8. Return both arms home and end the episode